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ORIGINAL RESEARCH

Net saving improvement of capacitor banks in power distribution systems by increasing daily size switching number: A comparative result analysis by artificial intelligence

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Abstract

This paper studies the effect of the number of switching (NOS) per day of capacitor banks on loss reduction in radial distribution systems. To this aim, the daytime (more precisely, 24 h) is divided into different numbers of time segments (equal to the same NOS) for capacitors' size switching. The resulting non-linear programming with discontinuous derivatives (called DNLP) model is solved subject to related constraints. The results reveal the impact of hourly switching of capacitor banks on further loss reduction (namely 118.4435, 83.7856, and 101.738 MWh for three IEEE systems) and higher net savings (i.e. k$5.6067, k$4.2772, and k$5.3542 for the same systems) of radial distribution systems compared to daily switching. Then, the hyper-tuned Random Forest model is trained based on the IEEE 69-bus network, fine-tuned by the IEEE 10-bus network, and fitted by the IEEE 33-bus network to have an intelligent multi-classification task with the highest accuracy. Numerical simulation, in both classic and intelligent parts, is presented to demonstrate the performance of DeepOptaCap. For the final step, DeepOptaCast is compared to other intelligent models of Light Gradient Boosting Method (LGBM), Decision Tree, and XGBoost, regarding KPIs of mean absolute percentage error, root mean squared percentage error, mean absolute error, root mean squared error, and coefficient of determination to demonstrate the model's superiority.

1 | INTRODUCTION

Capacitor banks are installed in distribution systems aiming at loss reduction by reactive power compensation [1] due to the rising importance of energy conservation in distribution systems [2]. They can also release the feeder capacity and improve the voltage profile as the other advantage of capacitor banks. The limitation factor for allocating capacitor banks in distribution systems is their investment cost [3]. For capacitor allocation, their lifecycle costs, including installation, O&M, and size (namely kVar) costs, must be considered for allocating capacitors in distribution systems, like other installation problems [4, 5]. Capacitor banks should reduce system energy loss, in addition to compensating for their lifecycle costs; otherwise, the capacitor placement is not economical.

Load demand varies with time within a 24-h horizon as well as different days of the year. Capacitor banks are associated with size-switching ability according to load demand variations for reactive compensation of electric networks [6]. By increasing the number of switching (NOS) per day, the system loss is further reduced [7]. In this regard, a 24-h period can be divided into different numbers of time segments (equal to the same NOS). Each time segment can have its own optimal size of capacitor (by capacitor steps) such that during a time segment, the capacitor size remains unchanged. Increasing NOS during a day results in decreased duration of time segments (i.e. higher accuracy); thus, the energy loss, the net saving, and the total cost are improved.

Contributing to the outlined context, the capacitor placement problem has been widely investigated in the literature. A

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capacitor placement problem to optimize the benefit of distribution companies has been suggested in [8]. Capacitor allocation in unbalanced systems has been investigated in [9–11], through a two-stage approach, for optimal location and optimal sizing, under loss minimization [9], using a hybrid honey bee colony algorithm [10], and based on using different load models under a penalty-free genetic algorithm [11]. Data clustering has been employed for capacitor allocation in wind-integrated distribution systems [12] and capacitor allocation via a clustering-based optimization approach [13]. A reliability-based capacitor placement problem in distribution systems considering inflation and interest rates has been proposed in [14] to show the impact of using unreal models on the total cost of distribution systems. In [15], the comparison between the results of the heuristic algorithms offered in the literature and the GAMS software has been accomplished for a proposed mixed-integer non-linear programming model of the capacitor allocation problem. A multi-objective capacitor placement problem for radial distribution systems has been presented in [16], in which an accelerated particle swarm optimization (PSO) was used to solve the problem. In another study [17], substation power factor improvement and loss minimization have been the objectives of capacitor placement in a radial distribution network. Grid voltage control has been focused on in another capacitor allocation problem in [6]. Different load levels have been considered for capacitor bank allocation based on the artificial bee colony algorithm [18]. Maximum net saving and enhanced system stability have been focused on for capacitor placement based on loss sensitivity factors [19, 20], in which the honey bee colony algorithm was adopted to determine optimal locations and for capacitor sizing. In [21], bio-inspired optimization approaches (including the bat algorithm and cuckoo search) have been adopted for loss reduction in a capacitor allocation problem aimed at cost-saving maximization. Power loss indexes have been employed for capacitor placement by the cuckoo search optimization algorithm [22] and the flower pollination optimization algorithm [23]. Additionally, a centralized system for compensating reactive power in distribution systems has been introduced in [24] by covering load growth in the future. The method was applied to a local shipyard power system. Moreover, in [25], a genetic algorithm has been utilized for the placement of capacitor banks with related control systems in radial networks.

A number of other works in this regard have focused on solving capacitor allocation problems by heuristic algorithms, including PSO algorithm [26, 27], genetic algorithms [28, 29], crow search algorithm [30], a hybrid method based on some heuristic methods [31], monkey search optimization technique [32], bacterial foraging optimization algorithm [33], teaching-learning algorithm [34], a hybrid optimization approach based on PSO and electromagnetic-like algorithms [35], and grasshopper optimization algorithm [36].

Moreover, in [37], a two-stage algorithm has been introduced for capacitor allocation. The first stage accomplished the sensitivity analysis to determine the location of capacitors, while in the second stage, the gravitational search algorithm performed the sizing of capacitors. Furthermore, capacitor placement for the Unified Egyptian Network, that is, the East Delta distribution Network has been accomplished in [38] under a two-stage approach. The first stage determined the optimal locations for installing capacitors, whereas the second stage using the ant colony approach accomplished the sizing of capacitor banks. In [7], the switching accuracy of a single capacitor bank from the energy loss minimization perspective has been evaluated for 1 year without economic considerations.

This study aims to extend the study accomplished in [7] by including economic considerations, namely the total costs of capacitors (the summation of the lifecycle cost and energy loss cost) and considering multiple capacitor banks (instead of one capacitor bank) under the lifespan of capacitor banks (instead of a single year). In addition, an optimization model is employed instead of using an iterative conventional algorithm. As mentioned in [7], the 24-h horizon has been divided into different time segments (equal to the same NOS) for optimal switching of capacitors, in which the size step of each capacitor bank changes at the beginning of each time segment. In this study, different levels of NOS for a day are evaluated from the viewpoint of lifecycle cost, energy loss cost, total costs, and net saving values, considering the problem constraints. To improve the complexity, first, the locations for installing capacitors are determined in the first stage. Then, the optimal base sizes of capacitor banks are optimized in the second stage. The size of each capacitor bank is its base size multiplied by the switching step (as a coefficient from 0 to 10). Different costs are taken into account for the lifecycle cost of capacitors, including the installation, O&M, and purchasing costs. For load flow calculations, the well-known backward–forward sweep approach is adopted. This approach is the fastest and most accurate technique for load flow in radial distribution systems. The resulting non-linear programming with discontinuous derivatives (DNLP) model is optimized using the PSO heuristic algorithm. Due to the easy implementation and tuning of PSO and the integer nature of the problem (all decision variables are integers), this algorithm ensures obtaining the global optimal solutions. Several test systems are adopted to investigate the effectiveness and generality of the suggested model. Finally, the artificial intelligence is employed to analyze the obtained results. This study is highlighted as follows:

- Analyzing different numbers of switching (namely NOS) per day from the viewpoint of energy loss, energy loss cost, lifecycle cost, total cost, net saving value, and the base size of capacitors.
- Adopting the PSO algorithm to optimize the resulting DNLP model.
- Load flow calculations using the backward–forward sweep method.
- Numerical simulation by adopting several radial distribution systems.
- Analyzing the results from the artificial intelligence perspective.

The remainder of this paper is structured as follows: Section 2 clarifies the NOS concept. The problem formulation is

presented in Section 3. The load flow calculations are explained in Section 4. In Section 5, the methodology and flowchart are discussed. Section 6 gives case studies to demonstrate the effectiveness and generality of the suggested model. The Artificial Intelligence to evaluate the obtained results is presented in Section 7. Finally, the conclusions of this work are described in Section 8.

2 | NOS CONCEPT

As mentioned in [7], for evaluating switching accuracy, a day is divided into several time segments. The duration of each time segment can be 1, 2, 3, 4, 6, 8, 12, and 24 h. For instance, for NOS = 3, time segment 1 is hours 1 to 8, time segment 2 is hours 9 to 16, and time segment 3 is hours 17 to 24. During each time segment, the size of capacitors remains unchanged, that is, the load flow is accomplished for all hours of a segment individually in the presence of an unchanged size of capacitors. It is clear that a higher NOS results in more loss improvement during a typical day (or longer time horizons), because by increasing NOS, the time segment duration (SD) is decreased, and capacitor banks can follow the load more accurately. Table 1 shows the division of a day into related time segments for different NOS values. Each segment has its own optimal size (1–10 multiplied by the base size of capacitor banks). Thus, NOS = 24 is the best case from the loss reduction perspective, since NOS = 24 results in $SD = 1$. The relation between these two parameters is depicted as

$$SD = 24/NOS \tag{1}$$

The start and end hours of each time segment (St_s and En_s, respectively) are as Equations (2) and (3), respectively, in which s illustrates the index of segments.

$$St_s = (s-1)SD + 1 \tag{2}$$

$$En_s = (s-1)SD + SD = sSD \tag{3}$$

The hours of each time segment for parametric NOS are shown in Table 2. As seen, for parametric s, the related time interval is $[(s-1)SD + 1,\ (s-1)SD + 2,\ \ldots,\ (s-1)SD + SD]$, which has identical optimal capacitor sizes. As mentioned, the capacitor banks' sizes are coefficients (from 1 to 10) of the base size for each capacitor bank. Additionally, the base size is a coefficient of 25 kVar. Therefore, each system capacitor has its optimal base size, and each time segment has its optimal size step from 1 to 10, which is multiplied by the base size.

3 | OBJECTIVE FUNCTION AND CONSTRAINTS

The intended problem is graphically outlined in Figure 1. As seen, the capacitor bank allocation problem is solved aiming at cost minimization. The total cost consists of two factors, includ-

TABLE 1 Switching intervals concerning the NOS.

NOS	Time (Hour)																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	$s=1$																							
2	$s=1$												$s=2$											
3	$s=1$								$s=2$								$s=3$							
4	$s=1$						$s=2$						$s=3$						$s=4$					
6	$s=1$				$s=2$				$s=3$				$s=4$				$s=5$				$s=6$			
8	$s=1$			$s=2$			$s=3$			$s=4$			$s=5$			$s=6$			$s=7$			$s=8$		
12	$s=1$		$s=2$		$s=3$		$s=4$		$s=5$		$t=6$		$s=7$		$s=8$		$s=9$		$s=10$		$s=11$		$s=12$	
24	$s=1$	$s=2$	$s=3$	$s=4$	$s=5$	$s=6$	$s=7$	$s=8$	$s=9$	$=10$	$=11$	$=12$	$=13$	$=14$	$=15$	$=16$	$=17$	$=18$	$=19$	$=20$	$=21$	$=22$	$=23$	$=24$

NOS, number of switching.

TABLE 2 Time segments for parametric NOS during a day.

Segments no.	Beginning of each time segment (in Hour)			
Segment 1	$t = 1$	$t = 2$	…	$t = SD$
Segment 2	$t = SD + 1$	$t = SD + 2$	…	$t = 2\,SD$
…	…	…	…	…
Segment s	$t = (s-1)\,SD + 1$	$t = (s-1)\,SD + 2$	…	$t = s\,SD$
…	…	…	…	…
Segment NOS	$t = (NOS-1)\,SD + 1$	$t = (NOS-1)\,SD + 2$	…	$t = NOS\,SD = 24$

NOS, number of switching.

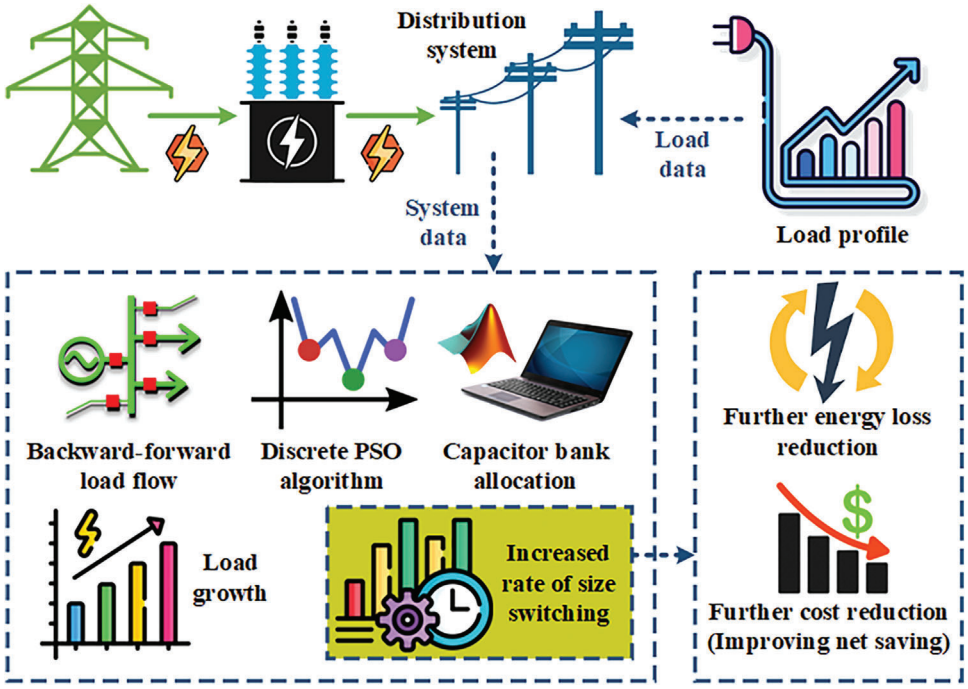


FIGURE 1 Diagram of the proposed capacitor bank allocation problem under the switching accuracy.

ing the lifecycle cost of capacitors as well as the system's energy loss during the lifespan of capacitors. Thus, the capacitor banks should at least compensate for the lifecycle cost by decreasing the loss cost. The lifecycle cost includes installation, O&M, and purchasing (namely kVar) costs. Since in this research, the capacitors have integer standard sizes (a factor of 25 kVar), as well as the presence of a capacitor in each bus is determined by a binary variable, a DNLP model is derived and optimized using the PSO algorithm. The difference between the decreased loss cost and the lifecycle cost shows the net saving of capacitors. The effect of the time accuracy in capacitors' switching on reducing network losses is evaluated under the NOS parameter (by dividing the 24-h horizon into different time segments), and the obtained results for different values of NOS are compared. It is clear that higher values of NOS yield lower energy loss and total cost. The power flow process is accomplished based on the well-known backward/forward sweep method, and the resulting model is optimized by the PSO algorithm. To improve the complicity of the model, the problem is optimized in two stages: the optimal location stage and the sizing stage. The sizing stage is to optimize the capacitors' base sizes and economic considerations. The second stage also optimizes capacitor switching (an integer variable from 0 to 10 multiplied by the base size of capacitors).

3.1 | Location optimization stage

In this stage, the location of capacitors is optimally determined to decrease the problem size. It is noted that among the obtained results of this stage, only the obtained locations (namely buses) are used as parameters in the sizing stage, and the other results of the first stage are ignored. Temporal variations (which is based on the load profile, monthly load levels, and yearly load growth) are considered in both this stage and the next stage.

To this aim, the variable N is defined, which shows the total number of buses that take a capacitor bank. For parametric N,

the index n is considered such that $1 < n < N$, and for each value of n, n buses are optimized. For instance, for $N = 1$, one location is optimized, for $N = 2$, two locations are optimized etc. The N is increased until the objective of the first stage starts to deteriorate. The N with the minimum loss is the optimal N, namely N_{opt}, and the set of the obtained buses (related to that N) shows the optimal locations, namely the result of this stage. Accordingly, the problem in the next stage has N_{opt} capacitor banks; thus, the second stage starts with these locations.

In the first stage, the size of capacitors is a natural coefficient multiplied by a standard capacitor size, namely 25 kVar as

$$Q_C^1(n) = 25\eta^n \quad \forall n \quad \& \quad \eta^n \in (1, 2, 3, ...) \tag{4}$$

in which $Q_C^1(n)$ stands for the capacitor size connected to the B_nth bus, and η^n represents the integer variable of the capacitors' size, as mentioned. Thus, the size of capacitor banks can be 25, 50, 75 etc., and in this stage, fixed integer sizes are considered for capacitor banks to determine the optimal locations.

To calculate the lifecycle cost of capacitors, the equivalent size of capacitors (namely $Size(n)$) is considered to better adapt to the next stage. This is because, in the next stage, each capacitor is a base size multiplied by a coefficient from 0 to 10. Thus, we have

$$Size(n) = 25(10x) \quad \forall n \tag{5}$$

$$\text{finding } x \quad \Rightarrow \quad 10(x-1) < \text{optimal}\,\eta^n \le 10x \quad \forall n \tag{6}$$

in which x is a natural number, that is, it can be 1, 2, 3 etc.

The cost function of the first stage, namely Equation (7), includes the energy loss cost in the presence of capacitor banks and the lifecycle costs of capacitor banks. The lifecycle costs include the investment cost and O&M cost. Also, the investment cost includes the purchasing cost and the installation cost. Thus, the lifecycle costs include the purchasing cost, the installation cost, and the O&M cost. The number of optimal locations is limited by the lifecycle cost since more than N_{Opt} capacitors is not economical, that is, the reduction in the energy loss is not limitless for a specific system. Therefore, the optimal number of capacitor banks is limited enough so that the capacitors can compensate for their lifecycle costs.

Problem I. min

$$TC^1 = \delta E_{Loss}^1 + \sum_n \left\{ Size(n) C_{Pu} + N_y C_{OM} + C_{Ins} \right\} \tag{7}$$

in which TC^1 stands for the total cost in the first stage, δ is the average electricity price during the understudy period, and E_{Loss}^1 represents the total energy loss during the planning period. Additionally, C_{Pu}, C_{OM}, and C_{Ins} are the purchasing cost (in \$/kVar), O&M cost (in \$/year), and installation cost (in \$) of the capacitor banks. It is clear that in Equation (7), the first term is the loss cost (i.e. C_{Loss}^1) and the second term is the lifecycle cost (i.e. C_{LCC}^1); therefore, $TC^1 = C_{Loss}^1 + C_{LCC}^1$. The investment cost can be limited as

$$C_{Inv}^1 \le C_{Inv}^{\max} \quad \& \quad C_{Inv}^1 = \sum_n \left\{ Size(n) C_{Pu} + C_{Ins} \right\} \tag{8}$$

where C_{Inv}^1 shows the investment cost in the first stage and C_{Inv}^{max} stands for the available capital for allocating capacitor banks. As mentioned, the first stage aims to determine N_{opt} and related locations. The other variables obtained in the first stage are ignored when the second stage starts.

3.2 | Sizing stage (main stage)

The cost function of this stage (i.e. the main stage) is expressed as (9). This cost function includes the loss cost in the presence of capacitor banks and the lifecycle costs of capacitor banks.

Problem II. min

$$TC = \delta E_{Loss} + \sum_n \left\{ N_{SW} Q_{Base}^n C_{Pu} + N_Y C_{OM} + C_{Ins} \right\} \tag{9}$$

in which TC stands for the total cost, Q_B^n shows the 'base' reactive power of the capacitor bank connected to the B_nth bus, E_{Loss} depicts the system's total energy loss during the planning period, and as mentioned, δ and N_y show the loss cost for a unit energy loss and planning period (in years), respectively. Also, N_{SW} illustrates the NOS steps (here 10). Therefore, $N_{SW} \times Q_{Base}^n = 10 \times Q_{Base}^n$ stands for the total size of the capacitor bank connected to the B_nth bus. It is clear that in Equation (9), the first term is the loss cost (i.e. C_{Loss}) and the second term is the lifecycle cost (i.e. C_{LCC}); therefore, $TC = C_{Loss} + C_{LCC}$. The investment cost is limited to the available initial capital as

$$C_{Inv} \le C_{Inv}^{Max} \quad \& \quad C_{Inv} = \sum_n \left\{ N_{SW} Q_{Base}(n) C_{Pu} + C_{Ins} \right\} \tag{10}$$

in which C_{Inv} shows the investment cost and as mentioned, C_{Inv}^{max} stands for the available capital for allocating capacitor banks.

In this research, the 'base' reactive power of the capacitor bank connected to the B_nth bus, namely Q_B^n, is an integer variable (namely α^n) multiplied by a standard size of 25 kVar as

$$Q_B^n = 25\alpha^n \quad \forall n = 1, 2, \ldots, N_{opt} \tag{11}$$

$$\alpha_n \in (1, 2, ...) \quad \forall n \tag{12}$$

The size of capacitor banks to be switched in different time segments is a switching coefficient (namely $\beta^{n,y,m,s}$) multiplied by Q_B^n as follows:

$$Q_{C,Seg}^{n,y,m,s} = \beta^{n,y,m,s} Q_B^n \quad \forall n, y, m, s \quad \& \quad \beta^{n,y,m,s} \in (0, 1, 2, ... 10) \tag{13}$$

Since the number of daily size switching of capacitor banks is limited to NOS, the capacitor size in the tth hour (namely $\beta^{n,y,m,t}$) is expressed as

$$Q_C^{n,y,m,t} = Q_{C,Seg}^{n,y,m,\left\lceil \frac{t}{SD} \right\rceil} \quad \forall n, y, m, t \tag{14}$$

in which '⌈ ⌉' stands for the ceiling function. The capacitors' reactive power can be limited to available capacitors in the market as

$$10\, Q_B^n \le Q_{C,\max}^n \quad \forall n, \forall t \tag{15}$$

The following equation is also adopted to limit the total reactive power compensation:

$$\sum_n Q_C^{n,y,m,t} \le \sum_i Q_L^{i,t} \quad \forall t \tag{16}$$

4 | LOAD FLOW CALCULATIONS

By considering the reactive power of the capacitor banks (namely $Q_C(\lambda(n))$, the active/reactive power injected into buses is as [39]

$$P(i) = P_L^i \forall i \tag{17}$$

$$Q(i) = Q_L^i \forall i \notin B_n \tag{18}$$

$$Q(B_n) = Q_L^{B_n} - Q_C(n) \forall i \in B_n \tag{19}$$

in which $P(i)$ and $Q(i)$ are active/reactive power injected into buses, respectively. In addition, P_L^i and Q_L^i are the active/reactive loads of the system.

To accomplish the power flow, the well-known backward/forward approach is adopted [40–42]. This approach is fast and accurate for the load flow process in radial distribution systems. Also, it can be easily implemented, which is explained in the following. First, $1\angle 0$ p.u. is assumed for the voltage of all buses (V_i^0, $\forall i$). The currents of the branches are accordingly obtained as follows:

$$I_i^k = \left(\frac{S_i}{V_i^{k-1}} \right)^* = \left(\frac{P(i) + jQ(i)}{V_i^{k-1}} \right)^* \tag{20}$$

where I_i^k and V_i^{k-1} represent the equivalent current injected into buses in the kth iteration and the equivalent voltage of buses in iteration $k+1$, respectively. In addition, S_i shows the complex power injected into the buses.

Backward step: Starting from terminal buses, the current of the system branches is obtained.

Forward step: Based on the obtained current for the system branches and the predetermined reference voltage, by starting from the reference bus, the voltage at the end of i to $i+1$ branch is calculated as follows:

$$V_{i+1}^k = V_i^k - I_{i+1}^k Z_{i+1} \tag{21}$$

in which I_{i+1} and Z_{i+1} are the current and impedance of the branches between buses i and $i+1$.

The total loss of the system is the summation of power losses in all branches [43, 44], which is obtained as

$$P_{Loss} = \sum_i^{N_b-1} \left| I_{i+1}^{Br} \right|^2 R_{i+1} \tag{22}$$

where I_{i+1}^{Br} and R_{i+1} are the current magnitude of the branches between buses i and $i+1$ and the resistance of the same branches. Thus, E_{Loss} is the summation of P_{Loss} during the planning period, when $Q_C^{n,y,m,t}$ is connected to the B_nth buses based on $Q_{C,Seg}^{n,y,m,s}$.

5 | METHODOLOGY

The flowchart of the suggested model to improve the net saving of capacitor banks based on increasing the daily size switching number is illustrated in Figure 2. The daily profile, monthly load levels, and yearly load growth are used to model the load in the period under study. The base size of each capacitor bank is a factor of 25 kVar, whereas the total capacitor size to be switched is a factor (an integer variable between 0 and 10) of the base size (namely 25 kVar). The impact of different numbers of switching in the day, namely NOS, on energy loss reduction is evaluated. The obtained results of the problem are optimal locations (namely buses), the base size of capacitors, the optimal switching, the optimal investment, and also energy loss reduction, energy loss cost reduction (namely gross saving), total cost reduction (namely net saving). As mentioned, the problem is solved in two stages: the location optimization stage and the sizing stage. The resulting model is optimized by the PSO algorithm.

6 | CASE STUDIES

The monthly load level of the first year is shown in Figure 3a, which is the same profile used in [45, 46]. Additionally, the yearly load level is illustrated in Figure 3b, in which a 5% annual growth in the load level is considered. Figure 3c depicts the load profile of the systems under study, namely the IEEE 10-bus, the IEEE 33-bus, and the IEEE 69-bus systems. The suggested problem is implemented in the MATLAB software and solved using the PSO algorithm. The proposed model gives various results, including energy loss, energy loss cost, lifecycle

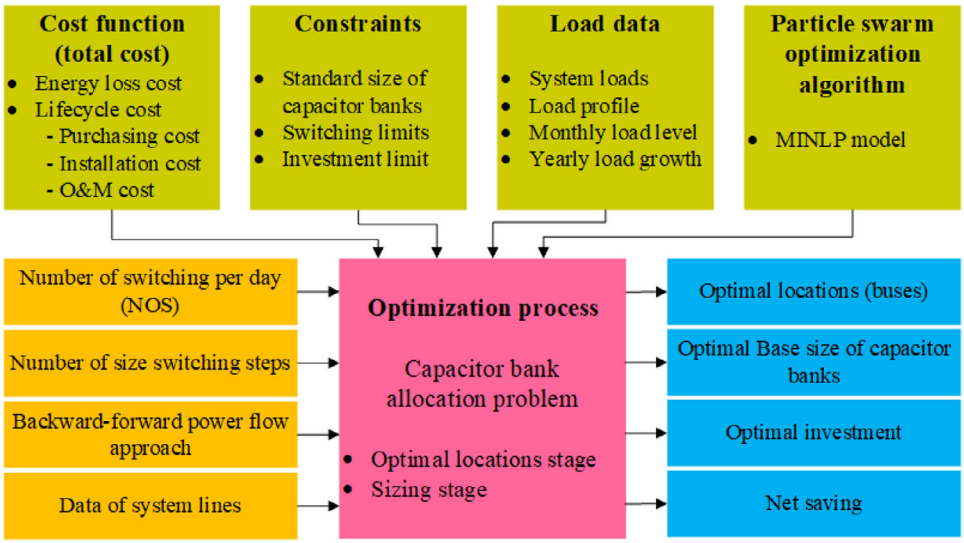


FIGURE 2 Flowchart of the offered model for capacitor bank allocation based on multi-period operation.

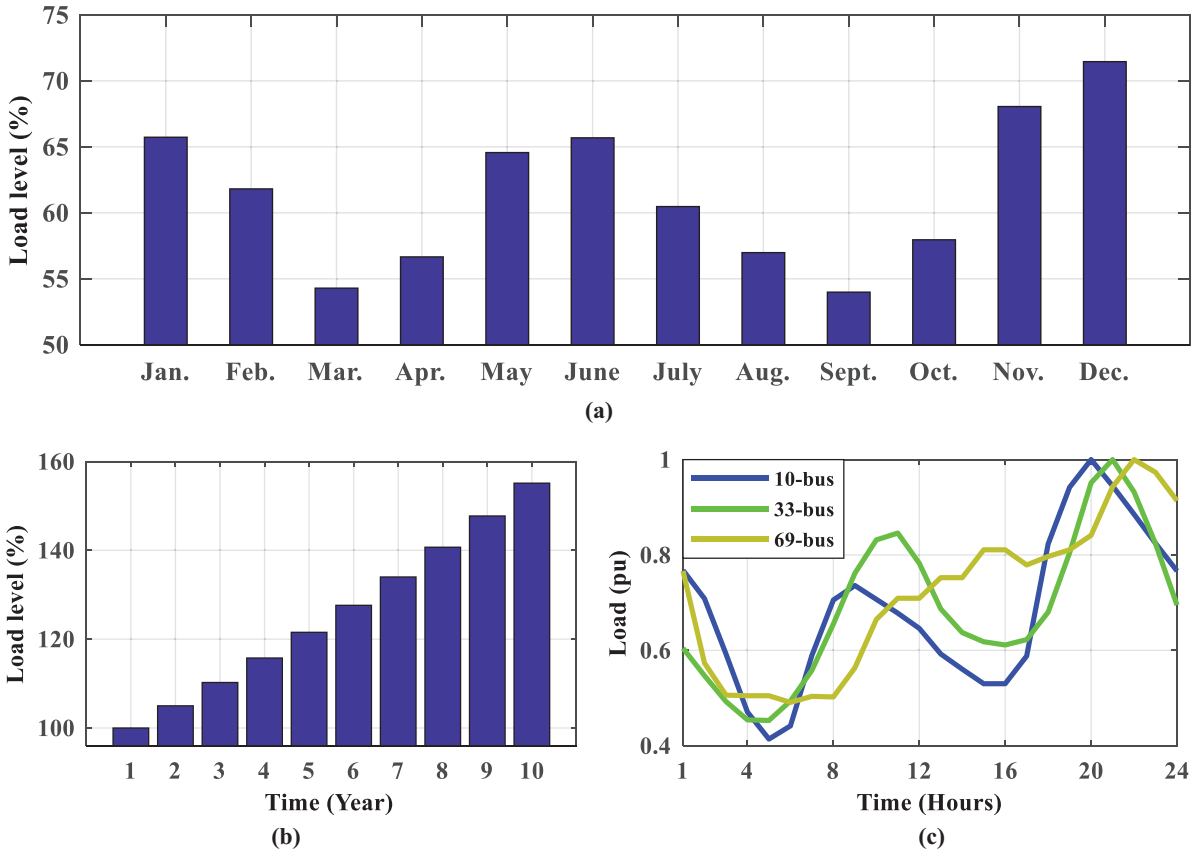


FIGURE 3 System load: (a) monthly load level, (b) yearly load level, (c) load profile of the systems under study.

cost, total cost (which is the cost function), net saving value, capacitor-integrated buses, and optimal size of capacitor banks. As mentioned, the base size for each capacitor bank is a factor of 25 kVar, and the switching coefficient is a discrete variable (from 0 to 10), which is multiplied by the base size. The base sizes of the capacitor banks are different and are obtained based on minimizing the cost function. In this study, the costs of unit energy loss, installation, O&M, and purchasing costs are considered \$0.06/kWh, \$1000, \$600/year, and \$3/kVar, respectively.

TABLE 3 Case I: Results of the location optimization stage for the IEEE 10-bus system.

N	Optimal buses	Sizes of the first stage [kVar]	Optimal Cost [$]	Optimal solution
0 (Base case)	–	–	1164589.3204	
1	5	2250	1089692.2971	
2	5, 9	1750, 350	1080788.5983	
3	5, 6, 10	1500, 500, 200	1079513.9508	✓
4	4, 5, 6, 10	500, 1250, 500, 200	1082268.0194	
5	3, 4, 5, 6, 10	1000, 500, 1000, 500, 225	1085885.9125	
6	3, 4, 5, 6, 8, 10	500, 500, 1000, 500, 175, 150	1089556.0209	
7	3, 4, 5, 6, 7, 8, 10	1000, 500, 1000, 250, 250, 125, 100	1093861.2086	
8	3, 4, 5, 6, 7, 8, 9, 10	500, 500, 1000, 250, 250, 75, 75, 125	1098311.1872	
9	2, 3, 4, 5, 6, 7, 8, 9, 10	250, 500, 500, 1000, 450, 100, 75, 75, 125	1103246.9213	

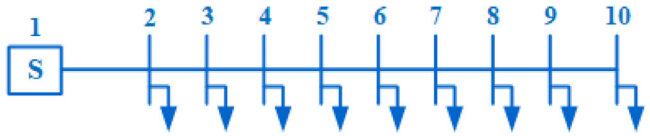


FIGURE 4 Case I: One-line schematic of the IEEE 10-bus system.

To justify the effectiveness of the suggested model, several small and large radial distribution systems, including IEEE 10-bus [47], 33-bus [48], and 69-bus [1] radial distribution networks, are employed as follows:

6.1 | Case I: IEEE 10-bus system

The IEEE 10-bus radial distribution system is adopted for allocating capacitor banks under the switching accuracy of capacitors. The single-line schematic of this 23-kV network is depicted in Figure 4. As mentioned, the optimal locations, namely the system buses, are determined in the first stage, whereas the sizing and economic operation of capacitor banks are studied in the second stage. Table 3 lists the optimal results of the location optimization stage. As mentioned, optimal buses are obtained by starting from the placement of one bus and increasing the number of buses until the loss deteriorates. As seen, buses 5, 6, and 10 are the most prioritized buses for allocating capacitors.

In the second stage, the sizing of capacitor banks (for the buses obtained in the first stage) is accomplished, that is, the obtained locations in the first stage are considered fixed in the second stage. This leads to decreased complexity in solving the second stage, which is the main stage of the problem. Table 4 depicts the results of the second stage, namely the sizing stage for the locations obtained in the previous stage (buses 5, 6, 10). Thus, this table does not include optimal buses. As seen, for $NOS = 1$, the energy loss is reduced from 19409.822 to 17679.3992 MWh (saving of 1730.4228 MWh), energy loss cost is improved from k$1164.5893 to k$1060.7640 (saving of k$103.8254), and the total cost is reduced from k$1164.5893 to k$1079.5140 (saving of k$85.0753).

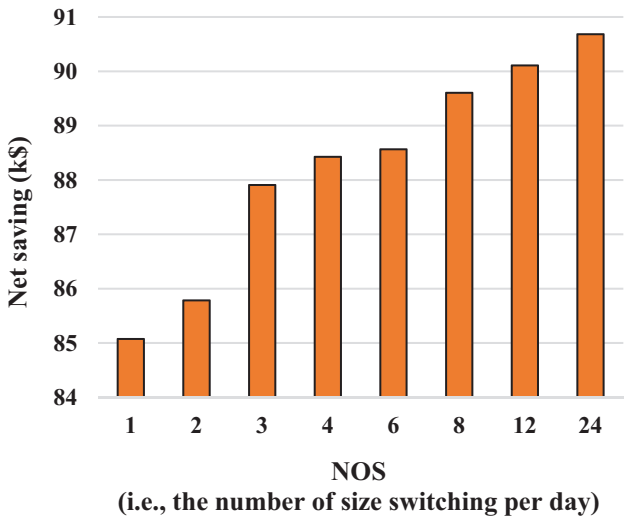


FIGURE 5 Case I: Net saving concerning NOS for the IEEE 10-bus system. NOS, number of switching.

On the other hand, (based on the novelty of this paper) for $NOS = 24$ (compared to $NOS = 1$), more improvements are achieved, namely 118.4435 MWh lower energy loss, k$7.1066 more gross saving (i.e., energy loss cost reduction), and k$5,6067 more net saving (i.e. total cost reduction) during the lifespan of capacitor banks. The latter improvements are related to $NOS = 24$ compared to $NOS = 1$, which is the main focus of this paper. The net saving concerning NOS is illustrated in Figure 5. As seen, the higher the NOS values, the greater the improvement in the net saving. It is worth mentioning that the net saving improvement trend (concerning NOS) depends on the related system as well as the daily load profile.

It is reminded that Tables 3 and 4 are the results of the two stages, respectively, for the IEEE 10-bus system, that is, Table 3 lists the results of the bus selection stage, whereas Table 4 shows the results of the sizing stage. The sizing and economic considerations are accomplished in the second stage (namely the sizing stage).

Table 5 lists the results for different NOS values compared to $NOS = 1$ for the IEEE 10-bus system. As seen, for $NOS =$

TABLE 4 Case I: Sizing and saving results for the IEEE 10-bus system.

Case	Lifecycle cost [k$]	Energy loss [MWh]	Energy loss cost [k$]	Total cost (lifecycle cost + energy loss cost) [k$]	Improved energy loss [MWh]	Improved loss cost (gross saving) [k$]	Improved total cost (net saving) [k$]	Base sizes [kVar] (buses: 5, 6, 10)
Base	–	19,409.822	1164.5893	1164.5893	–	–	–	–
$NOS = 1$	18.750	17,679.3992	1060.7640	1079.5140	1730.4228	103.8254	85.0753	150, 50, 25
$NOS = 2$	18.750	17,667.5774	1060.0546	1078.8046	1742.2446	104.5347	85.7847	150, 50, 25
$NOS = 3$	20.250	17,607.1925	1056.4316	1076.6816	1802.6295	108.1578	87.9077	175, 75, 25
$NOS = 4$	20.250	17,598.5551	1055.9133	1076.1633	1811.2669	108.6760	88.4260	175, 75, 25
$NOS = 6$	19.500	17,608.7504	1056.5250	1076.0250	1801.0717	108.0643	88.5643	150, 75, 25
$NOS = 8$	20.250	17,578.9206	1054.7352	1074.9852	1830.9014	109.8541	89.6041	175, 75, 25
$NOS = 12$	20.250	17,570.4915	1054.2295	1074.4795	1839.3305	110.3598	90.1098	175, 75, 25
$NOS = 24$	20.250	17,560.9557	1053.6573	1073.9073	1848.8663	110.9320	90.6820	175, 75, 25

NOS, number of switching.

TABLE 5 Case I: Gross and net saving concerning NOS for the IEEE 10-bus system.

	Improvement compared to $NOS = 1$	
Number of switching per day	Gross saving [k$]	Net saving [k$]
$NOS = 1$	–	–
$NOS = 2$	0.7093	0.7094
$NOS = 3$	4.3324	2.8324
$NOS = 4$	4.8506	3.3507
$NOS = 6$	4.2389	3.4890
$NOS = 8$	6.0287	4.5288
$NOS = 12$	6.5344	5.0345
$NOS = 24$	7.1066	5.6067

NOS, number of switching.

24, the gross saving and net saving are improved (i.e. increased) by k$7.1066 and k$5,6067, respectively. In other words, hourly switching of capacitors, compared to daily switching, results in k$5.6066 more net saving during the lifespan of capacitor banks.

6.2 | Case II: IEEE 33-bus system

As the second case study, the IEEE 33-bus radial distribution system is adopted for further evaluation of the suggested scheme. The single-line diagram of this 12.66-kV network is shown in Figure 6. The optimal results of the first stage are listed in Table 6, in which different values of N are evaluated until it is ensured that the least amount of TC (namely the total cost) is achieved; here $N = 3$ with the least cost. As can be seen for $N = 3$, buses 12, 24, and 30 are the buses with the most priorities for capacitor bank installation and thus are considered the optimal locations. The results of the second stage are presented in Table 7. As seen, the improvements for $NOS = 1$ are reducing the energy loss from 5394.6896 to 3679.0160 MWh (saving of 1715.6736 MWh) MWh, reducing the energy loss cost from k$323.6814 to k$220.7410 (gross saving of k$102.9404), and reducing the total cost from k$323.6814 to k$235.7410 (net saving of k$87.9404).

On the other hand, more improvements are achieved for $NOS = 24$, compared to $NOS = 1$, as of the considerable contribution of this work, namely 83.7856 MWh lower energy loss, as an increase from 1715.6736 to 1799.4592 MWh), k$5.0272 more gross saving (an increase from k$102.9404 to k$107.9676 in the loss cost), and k$4.2772 more net saving (an increase from k$87.9404 to k$92.2176 in the total cost) during the lifespan of capacitor banks. The latter improvements are related to $NOS = 24$ compared to $NOS = 1$, which demonstrates the effectiveness of hourly switching of capacitor banks in improving the economic operation of the IEEE 33-bus distribution system, similar to the previous test system. Figure 7 outlines the net saving concerning NOS. As this figure shows, the net saving is further improved for higher NOS values, namely more size switching of capacitor banks per day. As mentioned in Case I, the net saving improvement trend (concerning NOS) depends on the related system as well as the daily load profile.

It is reminded that Tables 6 and 7 are the results of the two stages, respectively, for the IEEE 33-bus system, that is, Table 6 lists the results of the bus selection stage, whereas Table 7 shows the results of the sizing stage. The sizing and economic considerations are accomplished in the second stage (i.e. the sizing stage).

Table 8 outlines the results for different NOS values compared to $NOS = 1$ for the IEEE 33-bus system. As seen, based on $NOS = 24$, the gross saving and net saving are improved (i.e., increased) by k$5.0272 and k$4.2772, respectively. Thus, hourly switching of capacitor banks (compared to daily switching) results in k$4.2772 more net saving during the lifespan of capacitor banks.

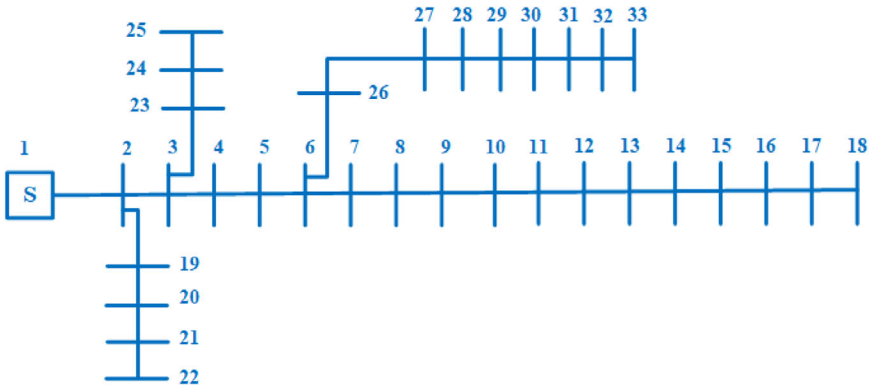


FIGURE 6 Case II: Single-line schematic of the IEEE 33-bus system.

TABLE 6 Case II: Results of the location optimization stage for the IEEE 33-bus system.

N	Optimal buses	Sizes of the first stage [kVar]	Optimal cost [$]	Optimal solution
0 (Base case)	–	–	323,681.3732	
1	30	700	242,970.7423	
2	12, 30	250, 575	236,454.8079	
3	12, 24, 30	250, 250, 500	235,740.9595	✓
4	7, 14, 24, 30	225, 150, 250, 500	237,689.6327	
5	7, 14, 24, 30, 32	225, 150, 250, 400, 100	241,892.4013	
6	7, 14, 21, 24, 30, 32	225, 150, 75, 250, 400, 100	246,292.6117	
7	7, 14, 21, 24, 29, 30, 32	225, 150, 75, 250, 125, 250, 125	250,367.4987	
8–32	–			

TABLE 7 Case II: Sizing and saving results for the IEEE 33-bus system.

Case	Lifecycle cost [k$]	Energy loss [MWh]	Energy loss cost [k$]	Total cost (lifecycle cost + energy loss cost) [k$]	Improved energy loss [MWh]	Improved loss cost (gross saving) [k$]	Improved total cost (net saving) [k$]	Base sizes (kVar) (buses: 12, 24, 30)
Base	–	5394.6896	323.6814	323.6814	–	–	–	–
NOS: 1	15.000	3679.0160	220.7410	235.7410	1715.6736	102.9404	87.9404	25, 25, 50
NOS: 2	15.750	3655.7569	219.3454	235.0954	1738.9327	104.3360	88.5860	25, 25, 75
NOS: 3	16.500	3618.5926	217.1156	233.6156	1776.0970	106.5658	90.0658	25, 50, 75
NOS: 4	15.750	3611.6512	216.6991	232.4491	1783.0384	106.9823	91.2323	25, 25, 75
NOS: 6	15.750	3613.8526	216.8312	232.5812	1780.8370	106.8502	91.1002	25, 25, 75
NOS: 8	15.750	3602.9691	216.1781	231.9281	1791.7205	107.5033	91.7533	25, 25, 75
NOS: 12	15.750	3598.3599	215.9016	231.6516	1796.3297	107.7798	92.0298	25, 25, 75
NOS: 24	15.750	3595.2304	215.7138	231.4638	1799.4592	107.9676	92.2176	25, 25, 75

6.3 | Case III: IEEE 69-bus system

As the third case study, the IEEE 69-bus radial distribution system is employed. The single-line diagram of this 12.66-kV system is illustrated in Figure 8. The optimal results for the first stage are listed in Table 9, in which different values of *N* are assessed from the total cost viewpoint. As seen, buses 17 and 50 have the most priorities for allocating capacitor banks; thus, they are considered the optimal locations. The optimal results of the second stage (based on the locations obtained in the first stage, namely buses 17 and 50) are presented in Table 10. As seen, the energy loss is reduced from 6360.9511 to 4315.7773 MWh (saving of 2045.1738 MWh), energy loss cost is improved from k$381.6571 to k$258.9466 (saving of

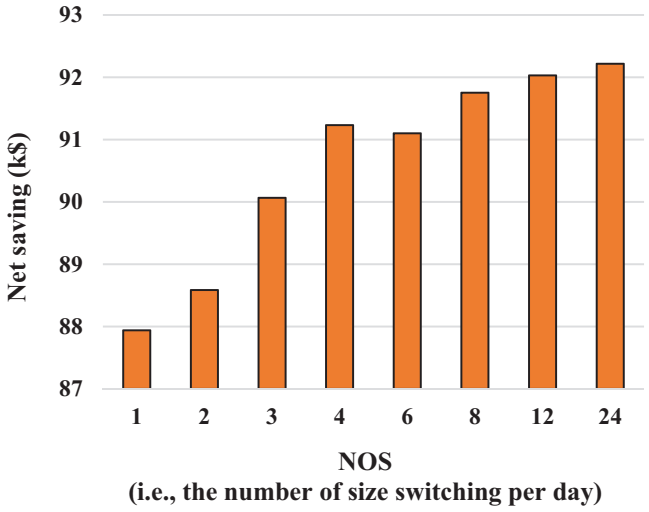


FIGURE 7 Case II: Net saving concerning NOS for the IEEE 33-bus system. NOS, number of switching.

TABLE 8 Case II: Gross and net saving concerning NOS for the IEEE 33-bus system.

Number of switching per day	Improvement compared to $NOS = 1$	
	Gross saving [k$]	Net saving [k$]
$NOS = 1$	–	–
$NOS = 2$	1.3956	0.6456
$NOS = 3$	3.6254	2.1254
$NOS = 4$	4.0419	3.2919
$NOS = 6$	3.9098	3.1598
$NOS = 8$	4.5629	3.8129
$NOS = 12$	4.8394	4.0894
$NOS = 24$	5.0272	4.2772

NOS, number of switching.

k$122.7105), and total cost is reduced from k$381.6571 to k$269.9466 (saving of k$111.7105).

For $NOS = 24$ more improvements are achieved compared to $NOS = 1$, namely 101.738 MWh further loss reduction. Also, more improvements of k$6.1042 and k$5.3542 in gross saving and net saving are achieved compared to $NOS = 1$. Thus, the effectiveness of higher values of *NOS* is proved again likewise 10-bus and 33-bus systems. The net saving concerning the NOS value is depicted in Figure 9. As seen, the higher the number of switching per day (namely the NOS values), the more improvement in the net saving amount. As mentioned in Cases I and II, the net saving improvement trend (concerning NOS) depends on the related system as well as the daily load profile.

It is reminded that Tables 9 and 10 are the results of the two stages, respectively, for the IEEE 69-bus system, that is, Table 9 lists the results of the bus selection stage, whereas Table 10 shows the results of the sizing stage. The sizing and economic considerations are accomplished in the second stage (i.e. the sizing stage).

Table 11 lists the results for different NOS values compared to $NOS = 1$ for the IEEE 69-bus system. As seen, based on $NOS = 24$, the gross saving and net saving are improved (i.e. increased) by k$6,1042 and k$5,3542, respectively. Thus, hourly switching of capacitor banks (compared to daily switching) results in k$4.2772 more net saving during the lifespan of capacitor banks.

In summary, the decision variables are optimal locations (namely buses) and optimal base sizes (which are multiplied by 0–10 for optimal switching) of capacitors. The other important results are optimal lifecycle cost, gross saving, and net saving. The optimal locations are outlined in Tables 3, 6, and 9. On the other hand, the optimal base sizes are outlined in Tables 4, 7, and 10. It is noted that Tables 4, 7, and 10 also show the economic results, that is, the lifecycle cost, gross saving, and net saving.

7 | FINAL STEP: DeepOptaCap

In this step, DeepOptaCap is utilized to perform the intelligent predictive classification task, as conceptualized in Figure 10. According to this figure, DeepOptaCap, a Random Forest model, is developed for accurate energy loss and loss cost forecasting in power systems. Trained on the IEEE 69 bus system, fine-tuned with data from the IEEE 10 bus system, and fitted to the IEEE 33 bus system, the model incorporates key features of *Optimal Investment [k$], Cost [k$], Improved Loss [MWh], and Improved Loss Cost [k$]*, predicting *Energy Loss [MWh], and Loss Cost [k$]*. KPIs are derived from the IEEE 33 bus system. As well, the intelligent structure of DeepOptaCast is illustrated in Figure 11.

Consequently, the proposed model is formulated as [49]

$$\psi_i = [OI\ IL\ LC\ EL\ C]_i \tag{23}$$

$$L = (\psi_i - \gamma)^2 \tag{24}$$

in which F_0, L, and ψ_i are the constant value of the prediction, loss function, and feature data. In terms of *arg min*, the optimization progress is expressed by

$$\frac{\partial}{\partial \gamma}\left(\sum_{i=1}^{M} L(\psi_i, \gamma)\right) = \frac{\partial}{\partial \gamma}\left(\sum_{i=1}^{M} (\psi_i - \gamma)^2\right) = -2\sum_{i=1}^{M} \psi_i$$

$$+2n\gamma - 2\sum_{i=1}^{M} P_i + 2n\gamma = 0 \rightarrow \gamma = \frac{1}{n}\sum_{i=1}^{M} \psi_i \tag{25}$$

Accordingly, the residual is obtained in each iteration as

$$F_0(x) = \psi_{Mean} = \frac{1}{n}\sum_{i=1}^{M} \psi_i$$

$$r_{im} = -\left[\frac{\partial L(\psi_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{M-1}(x)}$$

$$r_{im} = 2(\psi_i - F_{M-1}) \tag{26}$$

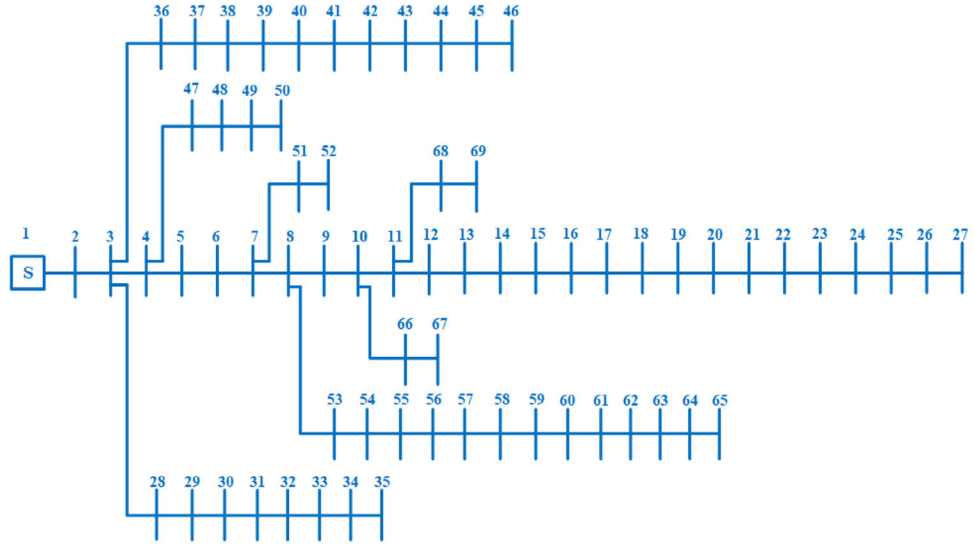


FIGURE 8 Case III: Single-line diagram of the IEEE 69-bus system.

TABLE 9 Case III: Results of the location optimization stage for the IEEE 69-bus system.

N	Optimal buses	Sizes of the first stage [kVar]	Optimal cost [$]	Optimal solution
0 (Base case)	–	–	381,657.0643	
1	50	750	274,140.0682	
2	17, 50	200, 725	269,839.6096	✓
3	18, 40, 50	175, 225, 700	273,287.3265	
4	18, 39, 40, 50	175, 250, 225, 700	276,737.7084	
5	12, 18, 39, 40, 50	150, 125, 250, 125, 700	280,582.9334	
6	12, 18, 39, 40, 50, 53	150, 125, 250, 150, 500, 170	284,216.3680	
7	11, 12, 18, 39, 40, 50, 53	100, 100, 125, 250, 125, 500, 175	288,829.9876	
8-68	–			

TABLE 10 Case III: Sizing and saving results for the IEEE 69-bus system.

Case	Lifecycle cost [k$]	Energy loss [MWh]	Energy loss cost [k$]	Total cost (lifecycle cost + energy loss cost) [k$]	Improved energy loss [MWh]	Improved loss cost (gross saving) [k$]	Improved total cost (net saving) [k$]	Base sizes (kVar) (buses: 17, 50)
Base	–	6360.9511	381.6571	381.6571	–	–	–	–
NOS: 1	11.000	4315.7773	258.9466	269.9466	2045.1738	122.7105	111.7105	25, 75
NOS: 2	11.000	4263.3582	255.8015	266.8015	2097.5929	125.8556	114.8556	25, 75
NOS: 3	11.750	4241.5030	254.4902	266.2402	2119.4481	127.1669	115.4169	25, 100
NOS: 4	11.750	4236.0895	254.1654	265.9154	2124.8616	127.4917	115.7417	25, 100
NOS: 6	11.750	4227.3829	253.643	265.393	2133.5682	128.0141	116.2641	25, 100
NOS: 8	11.750	4222.3715	253.3423	265.0923	2138.5796	128.3148	116.5648	25, 100
NOS: 12	11.750	4218.379	253.1027	264.8527	2142.5721	128.5544	116.8044	25, 100
NOS: 24	11750	4214.0393	252.8424	264.5924	2146.9118	128.8147	117.0647	25, 100

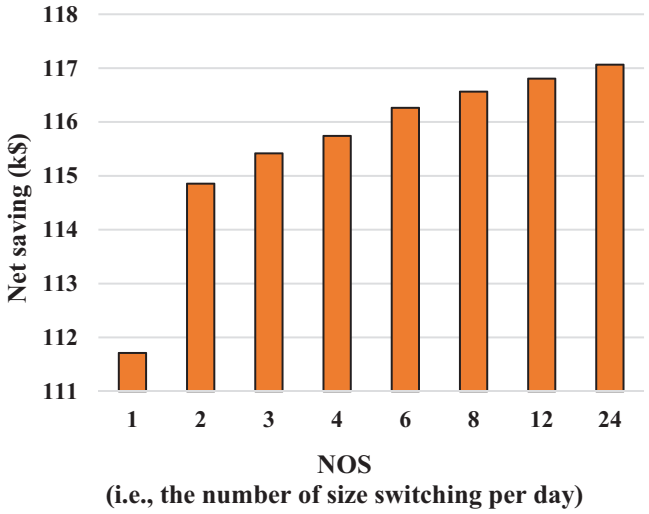


FIGURE 9 Case III: Net saving concerning NOS for the IEEE 69-bus system. NOS, number of switching.

TABLE 11 Case III: Gross and net saving concerning NOS for the IEEE 69-bus system.

Number of switching per day	Improvement compared to $NOS = 1$	
	Gross saving [k$]	Net saving [k$]
$NOS = 1$	–	–
$NOS = 2$	3.1451	3.1451
$NOS = 3$	4.4564	3.7064
$NOS = 4$	4.7812	4.0312
$NOS = 6$	5.3036	4.5536
$NOS = 8$	5.6043	4.8543
$NOS = 12$	5.8439	5.0939
$NOS = 24$	6.1042	5.3542

NOS, number of switching.

in which F_{M-1} is the last iteration of the prediction. The loss function optimization is performed by the terminal nodes as

$$\gamma_{im} = \arg\min \sum_{x_i \in R_{im}} L(\psi_i, F_{M-1}(x_i) + \gamma)$$

$$\gamma_{im} = \arg\min \sum_{x_i \in R_{im}} (\psi_i - F_{M-1}(x_i) - \gamma)^2 \quad (27)$$

$$\frac{\partial \gamma_{im}}{\partial \gamma} = 0 \rightarrow \frac{\partial}{\partial \gamma}\left[\sum_{x_i \in R_{im}} (\psi_i - F_{M-1}(x_i) - \gamma)^2\right] = 0$$

$$\gamma = \frac{1}{n_j} \sum_{x_i \in R_{im}} r_{im} \quad (28)$$

Then, the prediction is updated as

$$F_M(x) = F_{m-1}(x) + v \sum_{i=1}^{I_M} \gamma_{im} 1(x \in R_{im}) \quad (29)$$

As the last step, to evaluate the system performance, KPIs of mean absolute error (MAE), mean absolute percentage error (MAPE), root mean squared error (RMSE), root mean squared percentage error (RMSPE), and coefficient of determination (R^2) are utilized as [50, 51]

$$MAE = \frac{1}{n} \sum |(Y_{Observed} - Y_{Predicted})| \quad (30)$$

$$MAPE\ (\%) = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{Y_{i,Actual} - Y_{i,Predicted}}{Y_{i,Actual}} \right| \quad (31)$$

$$RMSE = \sqrt{\frac{1}{n} \sum (Y_{Observed} - Y_{Predicted})^2} \quad (32)$$

$$RMSPE\ (\%) = \sqrt{\frac{\sum_{i=1}^{n} (Y_{i,Actual} - Y_{i,Predicted})^2}{n \times Y_{i,Actual}}} \quad (33)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(Y_{i,Predicted} - Y_{i,Actual}\right)^2}{\sum_{i=1}^{n} \left(Y_{i,Actual} - \overline{Y}_i\right)^2} \quad (34)$$

in which the predicted and actual values are symbolized by $Y_{i,Predicted}$ and $Y_{i,Actual}$, respectively. The MAE is based on the mean absolute difference between the predicted/actual values, by providing a straightforward criterion for prediction accuracy. MAPE expresses errors as a percentage of the actual values, offering insight into the relative size of errors. RMSE is similar to MAE but with more weight to larger errors, providing an estimate of the spread of errors. RMSPE quantifies the square root of the mean of squared percentage errors, offering a percentage-wise assessment of error spread. The R^2 is based on the ratio of variance in the dependent variable captured by the model, where a higher R^2 indicates a better fit. As well, the results are available in Figure 12. These figures represent the density prediction profile, offering an insight into the distribution of predicted values for energy loss and cost loss, respectively. These density plots provide a visual representation of the predicted data distribution, aiding in understanding the central tendency and variability of the model predictions. In (c) and (d), the density residuals plots showcase the distribution of residuals, helping to assess the model's performance by visualizing the differences between the predicted and actual values. Finally, (e) and (f) illustrate the R^2 profile of energy loss and cost loss, respectively. These plots display the coefficient of determination (R^2) across different data points, offering a comprehensive view of how well the model explains the variability in energy loss and cost loss predictions. Moving to Figure 13, the KPI results of DeepOptaCap are presented. This figure likely encapsulates key performance indicators such as MAE, MAPE, RMSE, RMSPE, and R^2, providing a comprehensive evaluation of the model's accuracy and predictive capabilities. These KPIs serve as crucial metrics for assessing the overall performance and reliability

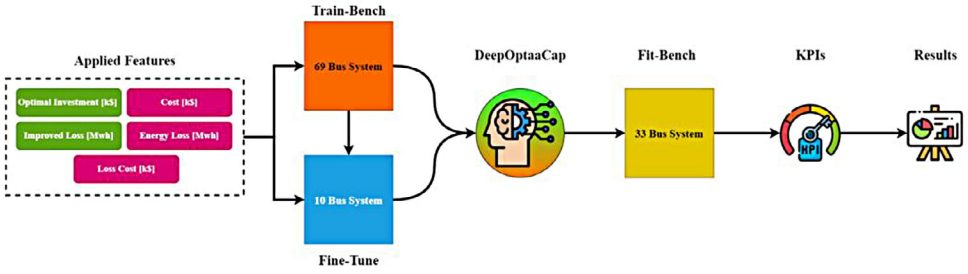


FIGURE 10 Overall procedure of DeepOptaCap.

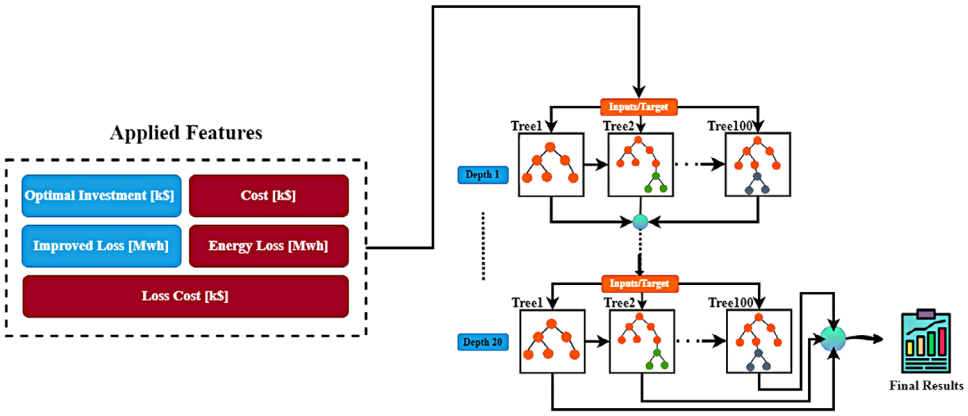


FIGURE 11 The structure of DeepOptaCap.

TABLE 12 Results of DeepOptaCast.

DeepOptaCap KPIs		
Energy loss	**MAPE**	0.45%
	RMSPE	1.28%
	RMSE	248.98
	MAE	86.21
	R^2	0.81
Loss cost	**MAPE**	0.45%
	RMSPE	1.28%
	RMSE	14.94
	MAE	5.17
	R^2	0.81

MAE, mean absolute error; MAPE, mean absolute percentage error; R^2, coefficient of determination; RMSE, root mean squared error; RMSPE, root mean squared percentage error.

of the DeepOptaCap model in predicting energy loss and cost loss. The results of Figure 13 are clarified in Table 12. Also, DeepOptaCap is compared with Light Gradient Boosting Model (LGBM), XGBoost, and Decision Tree, in terms of KPIs of MAPE, RMSPE, MAE, RMSE, and R^2, as available in Table 13 and Figure 14.

As the utilization of DeepOptaCap presented in NOS of capacitors banks, besides improving the application of artificial intelligence in this field, other several intelligent models can be applied to further considerations. Particularly, quantum technology [52] is one of these critical fields that investments are being devoted to its advanced and complex potentials [53]. One of the alternatives is quantum neural networks (QNNs) [54] that can be applied to forecasting, optimal planning, and classification fields of smart power systems, and capacitor banks-incorporated power distribution systems are not an exception. Consequently, for further developments, the role of quantum technology and its derivations can be taken into account in capacitor banks-incorporated power distribution systems.

8 | CONCLUSIONS

In this study, the switching accuracy of capacitor banks in radial distribution systems was evaluated from the energy and net saving considerations. To this aim, a two-stage strategy was presented, including the location optimization stage and the sizing stage. The optimization objective, namely the cost function, included the lifecycle cost of capacitor banks (including installation, O&M, and purchasing costs), as well as the network's energy loss cost. The model was optimized for different daily size switching numbers (namely the same NOS). Furthermore, the power flow process was performed based on the

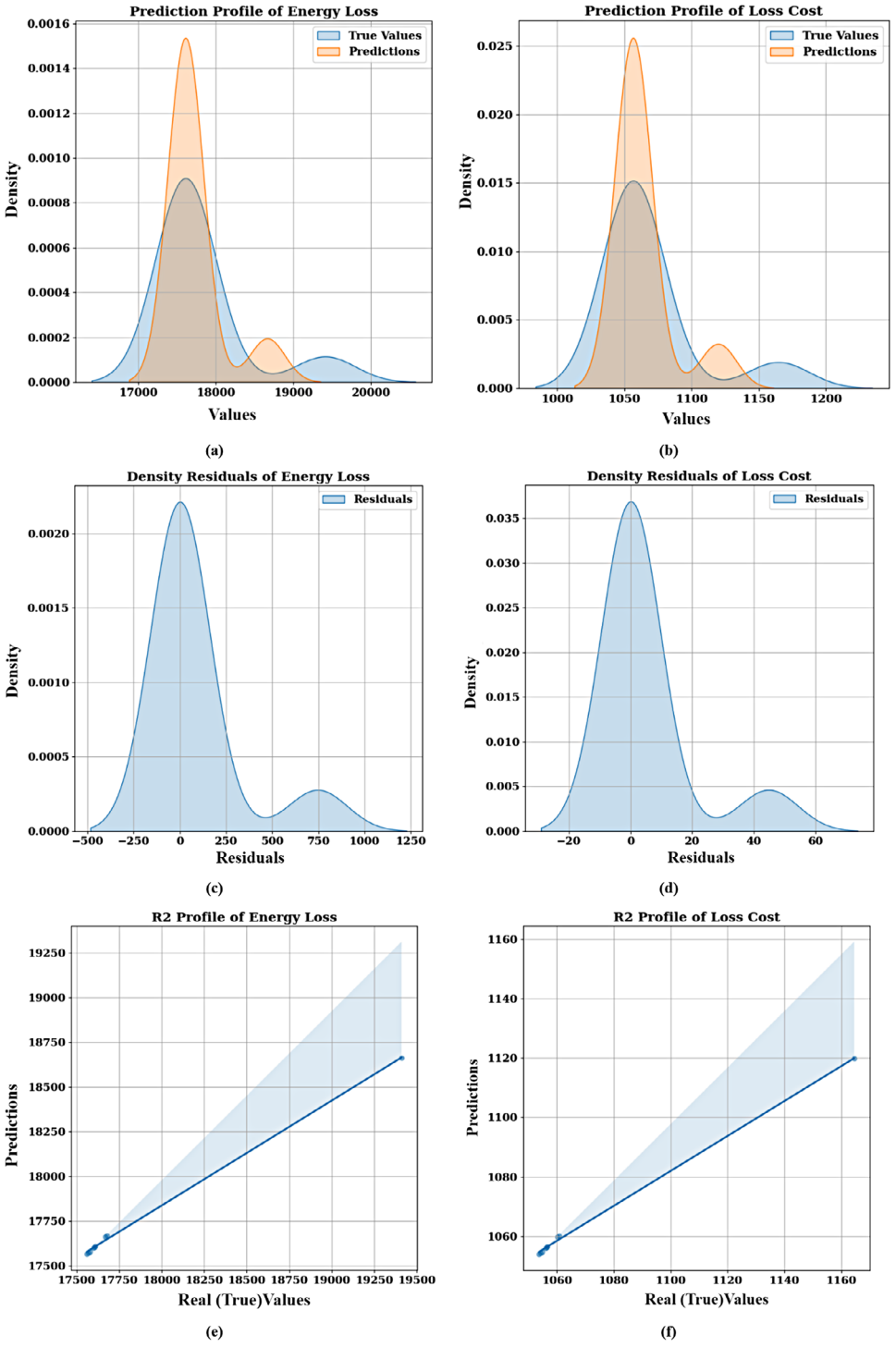


FIGURE 12 (a, b) Density prediction profile, (c, d) density residuals, and (e, f) R^2 profile of energy loss, and cost loss, respectively.

well-known backward/forward sweep method. The problem was formulated as a DNLP model and solved using the PSO algorithm. The numerical simulation was presented by adopting numerous radial distribution systems (including IEEE 10-bus, IEEE 33-bus, and IEEE 69-bus systems) to depict the impact of increasing NOS on energy loss, overall cost, and net saving values. Higher values of NOS resulted in lower costs and energy losses. The findings indicated further loss reduction, namely 118.4435, 83.7856, and 101.738 MWh for three IEEE systems, and higher net savings, that is, k$5.6067, k$4.2772, and k$5.3542 for the same systems, for hourly switching in radial distribution systems compared to daily switching.

TABLE 13 KPI comparisons.

Model	MAE energy loss	RMSE energy loss	R^2 energy loss	MAE loss cost	RMSE loss cost	R^2 loss cost
DeepOptaCap	61.33	61.33	0.81	3.68	3.68	0.81
LGBM	222.8	222.8	0	13.37	13.37	0
XGBoost	52.42	52.42	0	3.14	3.14	0
Decision tree	52.42	52.42	0	3.15	3.15	0

LGBM, light gradient boosting method; MAE, mean absolute error; MAPE, mean absolute percentage error; R^2, coefficient of determination; RMSE, root mean squared error; RMSPE, root mean squared percentage error.

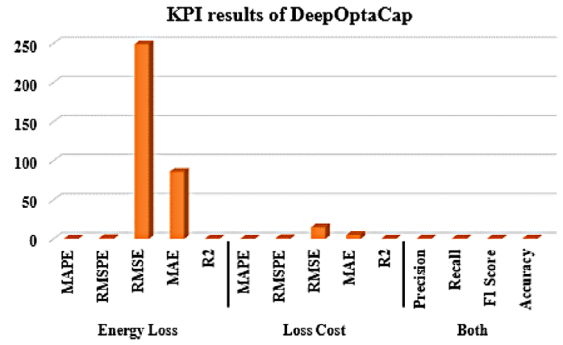


FIGURE 13 KPI results of DeepOptaCap.

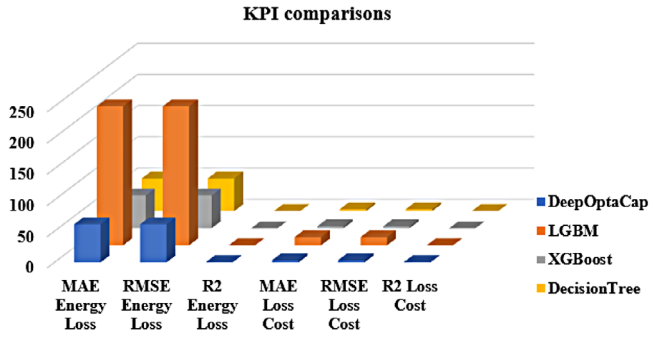


FIGURE 14 KPI comparison results.

NOMENCLATURE

Indexes

i index of the system buses
t index of hours
s index of time segments is a day
n index of the locations of capacitor banks
k index of iterations in backward-forward sweep load flow
y index of years
m index of months

Parameters and scalars

NOS number of switching (during a day)
SD time segment duration
St_s start hour of segments
En_s end hour of segments
N parameter for the total number of locations that take capacitor banks
N_{Opt} optimal number of buses/locations for installing capacitor banks
N_{SW} number of switching steps
N_y lifespan of capacitors or so-called planning period [year]
C_{Pu} purchasing cost of capacitors [$/kVar]
C_{OM} O&M cost of one capacitor bank [$/year]
C_{Ins} installation cost of one capacitor bank [$]
δ average electricity price [$/kWh]
C_{Inv}^{max} available capital for allocating capacitor banks [$]
$P(i)$ injection of active power into system buses
$S(i)$ injection of complex power into system buses
P_L^i active loads
Q_L^i reactive loads
R_{i+1} resistance of system branches
Z_{i+1} impedance of system branches

Variables

B_n optimal bus related to index n
α^n integer variable of capacitors' base sizes
$\beta^{n,y,m,s}$ integer variable for switching of capacitor banks
TC total cost [$]
E_{Loss} system energy losses during the planning period [MWh]
C_{Loss} energy loss cost [$]
C_{Inv} investment cost (i.e. purchasing and installation costs) of capacitor banks [$]
C_{LCC} lifecycle cost of capacitor banks [$]
$Q_{Base}(n)$ base size of capacitor banks [kVaR]
$Q_{C,Seg}^{n,y,m,s}$ capacitors' reactive power in sth time segment [kVaR]
$Q_C^{n,y,m,t}$ capacitors' reactive power in tth hour
$Q(i)$ injection of reactive power into system buses
$Q_C(i)$ reactive power of capacitors in the buses
V_i^k equivalent voltage of ith bus in iteration k
I_i^k equivalent current of ith bus in iteration k
I_{i+1}^{Br} current of system branches

AUTHOR CONTRIBUTIONS

Omid Sadeghian: Investigation; methodology; resources; software; writing—original draft. **Ashkan Safari**: Conceptualization; formal analysis; validation; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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DATA AVAILABILITY STATEMENT

Data will be made available on request.

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REFERENCES

1. Baran, M.E., Wu, F.F.: Optimal capacitor placement on radial distribution systems. IEEE Trans. Power Deliv. 4(1), 725–734 (1989)
2. Sadeghian, O., Mohammadi-ivatloo, B., Oshnoei, A., Aghaei, J.: Unveiling the potential of renewable energy and battery utilization in real-world public lighting systems : A review. Renew. Sustain. Energy Rev. 192, 114241 (2024)
3. Kalambe, S., Agnihotri, G.: Loss minimization techniques used in distribution network: Bibliographical survey. Renew. Sustain. Energy Rev. 29, 184–200 (2014)
4. Li, Y., Feng, B., Li, G., Qi, J., Zhao, D., Mu, Y.: Optimal distributed generation planning in active distribution networks considering integration of energy storage. Appl. Energy 210, 1073–1081 (2018)
5. Li, Y., Feng, B., Wang, B., Sun, S.: Joint planning of distributed generations and energy storage in active distribution networks: A bi-level programming approach. Energy 245, 123226 (2022)
6. Homaee, O., Zakariazadeh, A., Jadid, S.: Real-time voltage control algorithm with switched capacitors in smart distribution system in presence of renewable generations. Int. J. Electr. Power Energy Syst. 54, 187–197 (2014)
7. Sadeghian, O., Oshnoei, A., Kheradmandi, M., Mohammadi-Ivatloo, B.: Optimal placement of multi-period-based switched capacitor in radial distribution systems. Comput. Electr. Eng. 82, 106549 (2020)
8. Karimi, H., Dashti, R.: Comprehensive framework for capacitor placement in distribution networks from the perspective of distribution system management in a restructured environment. Int. J. Electr. Power Energy Syst. 82, 11–18 (2016)
9. de Araujo, L.R., Penido, D.R.R., Carneiro, S., Pereira, J.L.R.: Optimal unbalanced capacitor placement in distribution systems for voltage control and energy losses minimization. Electr. Power Syst. Res. 154, 110–121 (2018)
10. Taher, S.A., Bagherpour, R.: A new approach for optimal capacitor placement and sizing in unbalanced distorted distribution systems using hybrid honey bee colony algorithm. Int. J. Electr. Power Energy Syst. 49(1), 430–448 (2013)
11. Vuletić, J., Todorovski, M.: Optimal capacitor placement in distorted distribution networks with different load models using Penalty Free Genetic Algorithm. Int. J. Electr. Power Energy Syst. 78, 174–182 (2016)
12. Sadeghian, O., Oshnoei, A., Khezri, R., Hagh, M.T.: Data clustering-based approach for optimal capacitor allocation in distribution systems including wind farms. IET Gener. Transm. Distrib. 13(15), 3397–3408 (2019)
13. Vuletić, J., Todorovski, M.: Optimal capacitor placement in radial distribution systems using clustering based optimization. Int. J. Electr. Power Energy Syst. 62, 229–236 (2014)
14. Rahmani-Andebili, M.: Reliability and economic-driven switchable capacitor placement in distribution network. IET Gener. Transm. Distrib. 9(13), 1572–1579 (2015)
15. Nojavan, S., Jalali, M., Zare, K.: Optimal allocation of capacitors in radial/mesh distribution systems using mixed integer nonlinear programming approach. Electr. Power Syst. Res. 107, 119–124 (2014)
16. El-Fergany, A.A., Abdelaziz, A.Y.: Efficient heuristic-based approach for multi-objective capacitor allocation in radial distribution networks. IET Gener. Transm. Distrib. 8(1), 70–80 (2014)
17. Gampa, S.R., Das, D.: Optimum placement of shunt capacitors in a radial distribution system for substation power factor improvement using fuzzy GA method. Int. J. Electr. Power Energy Syst. 77, 314–326 (2016)
18. Dixit, M., Kundu, P., Jariwala, H.R.: Optimal integration of shunt capacitor banks in distribution networks for assessment of techno-economic asset. Comput. Electr. Eng. 71, 331–345 (2018)
19. El-Fergany, A.A., Abdelaziz, A.Y.: Capacitor placement for net saving maximization and system stability enhancement in distribution networks using artificial bee colony-based approach. Int. J. Electr. Power Energy Syst. 54, 235–243 (2014)
20. El-Fergany, A.A.: Involvement of cost savings and voltage stability indices in optimal capacitor allocation in radial distribution networks using artificial bee colony algorithm. Int. J. Electr. Power Energy Syst. 62, 608–616 (2014)
21. Injeti, S.K., Thunuguntla, V.K., Shareef, M.: Optimal allocation of capacitor banks in radial distribution systems for minimization of real power loss and maximization of network savings using bio-inspired optimization algorithms. Int. J. Electr. Power Energy Syst. 69, 441–455 (2015)
22. El-Fergany, A.A., Abdelaziz, A.Y.: Capacitor allocations in radial distribution networks using cuckoo search algorithm. IET Gener. Transm. Distrib. 8(2), 223–232 (2014)
23. Abdelaziz, A.Y., Ali, E.S., Abd Elazim, S.M.: Flower pollination algorithm and loss sensitivity factors for optimal sizing and placement of capacitors in radial distribution systems. Int. J. Electr. Power Energy Syst. 78, 207–214 (2016)
24. Chen, S.X., Eddy, Y.S.F., Gooi, H.B., Wang, M.Q., Lu, S.F.: A centralized reactive power compensation system for LV distribution networks. IEEE Trans. Power Syst. 30(1), 274–284 (2015)
25. Moradian, S., Homaee, O., Jadid, S., Siano, P.: Optimal placement of switched capacitors equipped with stand-alone voltage control systems in radial distribution networks. Int. Trans. Electr. Energy Syst. 29(3), e2753 (2019)
26. Lee, C.S., Ayala, H.V.H., Coelho, L.D.S.: Capacitor placement of distribution systems using particle swarm optimization approaches. Int. J. Electr. Power Energy Syst. 64, 839–851 (2015)
27. Rezaee Jordehi, A., Jasni, J.: Particle swarm optimisation for discrete optimisation problems: a review. Artif. Intell. Rev. 43(2), 243–258 (2012)
28. Esmaeili, D., Zare, K., Mohammadi-Ivatloo, B., Nojavan, S.: Simultaneous optimal network reconfiguration, DG and fixed/switched capacitor banks placement in distribution systems using dedicated genetic algorithm. Majlesi J. Electr. Eng. 9(4), 2015 (2015)
29. Jafari, A., Ganjeh Ganjehlou, H., Khalili, T., Mohammadi-Ivatloo, B., Bidram, A., Siano, P.: A two-loop hybrid method for optimal placement and scheduling of switched capacitors in distribution networks. IEEE Access 8, 38892–38906 (2020)
30. Askarzadeh, A.: Capacitor placement in distribution systems for power loss reduction and voltage improvement: A new methodology. IET Gener. Transm. Distrib. 10(14), 3631–3638 (2016)
31. Chiou, J.-P., Chang, C.-F.: Development of a novel algorithm for optimal capacitor placement in distribution systems. Int. J. Electr. Power Energy Syst. 73, 684–690 (2015)
32. Duque, F.G., De Oliveira, L.W., De Oliveira, E.J., Marcato, A.L.M., Silva, I.C.: Allocation of capacitor banks in distribution systems through a modified monkey search optimization technique. Int. J. Electr. Power Energy Syst. 73, 420–432 (2015)
33. Devabalaji, K.R., Ravi, K., Kothari, D.P.: Optimal location and sizing of capacitor placement in radial distribution system using Bacterial Foraging Optimization Algorithm. Int. J. Electr. Power Energy Syst. 71, 383–390 (2015)
34. Sultana, S., Roy, P.K.: Optimal capacitor placement in radial distribution systems using teaching learning based optimization. Int. J. Electr. Power Energy Syst. 54, 387–398 (2014)
35. Shefaei, A., Vahid-Pakdel, M.J., Mohammadi-ivatloo, B.: Application of a hybrid evolutionary algorithm on reactive power compensation problem of distribution network. Comput. Electr. Eng. 72, 125–136 (2018)
36. Elsayed, A.M., Mishref, M.M., Farrag, S.M.: Optimal allocation and control of fixed and switched capacitor banks on distribution systems using

grasshopper optimisation algorithm with power loss sensitivity and rough set theory. IET Gener. Transm. Distrib. 13(17), 3863–3878 (2019)
37. Mohamed Shuaib, Y., Surya Kalavathi, M., Christober Asir Rajan, C.: Optimal capacitor placement in radial distribution system using gravitational search algorithm. Int. J. Electr. Power Energy Syst. 64, 384–397 (2015)
38. Abou El-Ela, A.A., El-Sehiemy, R.A., Kinawy, A.M., Mouwafi, M.T.: Optimal capacitor placement in distribution systems for power loss reduction and voltage profile improvement. IET Gener. Transm. Distrib. 10(5), 1209–1221 (2016)
39. Sadeghian, O., Oshnoei, A., Tarafdar-Hagh, M., Khezri, R.: A clustering-based technoeconomic analysis for wind farm and shunt capacitor allocation in radial distribution systems. Int. Trans. Electr. Energy Syst. 31(1), e12708 (2021)
40. Shirmohammadi, D., Hong, H.W., Semlyen, A., Luo, G.X.: A compensation-based power flow method for weakly meshed distribution and transmission networks. IEEE Trans. Power Syst. 3(2), 753–762 (1988)
41. Sadeghian, O., Oshnoei, A., Kheradmandi, M., Khezri, R., Mohammadi-Ivatloo, B.: A robust data clustering method for probabilistic load flow in wind integrated radial distribution networks. Int. J. Electr. Power Energy Syst. 115, 105392 (2020)
42. Sadeghian, O., Oshnoei, A., Tarafdar-Hagh, M., Kheradmandi, M.: A clustering-based approach for wind farm placement in radial distribution systems considering wake effect and a time-acceleration constraint. IEEE Syst. J. 15(1), 985–995 (2021)
43. Li, Y., Han, M., Shahidehpour, M., Li, J., Long, C.: Data-driven distributionally robust scheduling of community integrated energy systems with uncertain renewable generations considering integrated demand response. Appl. Energy 335, 120749 (2023)
44. Li, Y., Wang, J., Zhao, D., Li, G., Chen, C.: A two-stage approach for combined heat and power economic emission dispatch: Combining multi-objective optimization with integrated decision making. Energy 162, 237–254 (2018)
45. Sadeghian, O., Mohammadpour Shotorbani, A., Mohammadi-Ivatloo, B., Ghassemzadeh, S.: Fuel cell preventive maintenance in an electricity market with hydrogen storage and scenario-based risk management. Sustain. Energy Technol. Assessments 61, 103587 (2024)
46. Sadeghian, O., Mohammadpour Shotorbani, A., Mohammadi-Ivatloo, B., Sadiq, R., Hewage, K.: Risk-averse maintenance scheduling of generation units in combined heat and power systems with demand response. Reliab. Eng. Syst. Saf. 216, 107960 (2021)
47. Baghzouz, Y., Ertem, S.: Shunt capacitor sizing for radial distribution feeders with distorted substation voltages. IEEE Trans. Power Deliv. 5(2), 650–657 (1990)
48. Baran, M.E., Wu, F.F.: Network reconfiguration in distribution systems for loss reduction and load balancing. IEEE Trans. Power Deliv. 4(2), 1401–1407 (1989)
49. Safari, A., Gharehbagh, H.K., Nazari-Heris, M., Oshnoei, A.: DeepResTrade: A peer-to-peer LSTM-decision tree-based price prediction and blockchain-enhanced trading system for renewable energy decentralized markets. Front. Energy Res. 11, 1275686 (2023)
50. Safari, A., Kheirandish Gharehbagh, H., Nazari Heris, M.: DeepVELOX: INVELOX wind turbine intelligent power forecasting using hybrid GWO–GBR algorithm. Energies 16(19), 6889 (2023)
51. Safari, A., Kharrati, H., Rahimi, A.: Multi-term electrical load forecasting of smart cities using a new hybrid highly accurate neural network-based predictive model. Smart Grids Sustain. Energy 9(1) (2024)
52. Ashkan Safari, A.A.G.: Use case of artificial intelligence, and neural networks in energy consumption markets, and industrial demand response. In: Industrial Demand Response: Methods, Best Practices, Case Studies, and Applications, IET, p. 379 (2022)
53. Safari, A., Ghavifekr, A.A.: Quantum neural networks (QNN) application in weather prediction of smart grids. In: 2021 11th Smart Grid Conference (SGC) (2021)
54. Safari, A., Ghavifekr, A.A.: Quantum technology quantum neural networks in smart grids control: Premier perspectives. In: 2022 8th International Conference on Control, Instrumentation and Automation, ICCIA 2022 (2022)

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